An upstream placebo shows that reservoir siting explains apparent drought buffering during Chile's megadrought

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Much of what is attributed to reservoir drought buffering may be a measurable siting confound. We treat reservoir presence as a basin-level intervention in Chile's megadrought and separate siting from operation with an upstream/downstream placebo, comparing regulated reaches below each dam against unregulated reaches above. Across streamflow, the apparent buffering is as strong upstream as downstream (slopes -0.20 versus -0.16; pooled gap $p = 0.39$, paired contrast $p = 0.72$). Hard-balancing aridity (standardized mean difference 0.000, effective control 19) makes even the apparent buffering vanish (intent-to-treat -0.02, $p > 0.85$). Over the streamflow record, the first decade of the megadrought ending March 2020 and excluding the 2019–2022 peak, the null is well powered, excluding only large operational buffering (above 46 to 59% of baseline; moderate buffering remains possible), and a positive control recovers injected effects one to one. Dammed basins hold 4.5 times more cropland than matched controls, a siting legacy with no detected divergence within power. The storage band drifts downward, shrinking the carryover available to buffer multi-year drought. Cropland expansion and water-rights accrual are consistent with no effect but underpowered; evapotranspiration is excluded by a documented product bias. We detect no operational buffering within the design's bounds and no megadrought-era demand expansion beyond the siting legacy. The upstream placebo is a portable method for regulated rivers. Whether the demand-side null generalizes, or marks a fully allocated market, remains to be tested. Adaptation must shift toward managing demand.

1 Plain Language Summary

When drought strikes, governments often respond by building more reservoirs, on the assumption that storing water cushions communities when rainfall fails. But dams are built where water is already scarce, so basins with dams differ from basins without them even before any drought. We separated the effect of the dam from the effect of its location by comparing river gauges above and below each dam, and by matching dry dammed basins to similar undammed ones across Chile's decade long megadrought. Once location is accounted for, reservoirs show no measurable buffer against multi year drought: streamflow, farmland, and water rights all respond the same with or without a dam, and the apparent buffering reflects where dams are built, not what they do. With the water available to store falling, our results suggest that building more reservoirs is not the answer; managing how water is used is.

2 Introduction

A persistent obstacle separates what we think reservoirs do during drought from what they actually do: dams are built where water is already scarce, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit of storage is documented (Wu et al. 2017, 2018), and model-based counterfactuals, in which a reservoir's effect is read against reconstructed or simulated unregulated flow, find that regulation attenuates hydrological drought downstream (Wu et al. 2019, 2021; Wan et al. 2017). Whether this buffering extends to multi-year drought, however, remains contested, because those counterfactuals are themselves model-dependent: they can embed assumptions about where the dam sits and about the unregulated baseline it is measured against. The distinction matters. If buffering is real, expanding storage is a rational adaptation to intensifying drought. If it is siting, then the reflexive global response, more dams, invests in the wrong mechanism.

The reservoir-effect hypothesis (Di Baldassarre et al. 2018) formalizes the concern: greater supply invites expansion, so demand grows to meet and then exceed each increment of storage (the supply-demand cycle), and prolonged reliance on stored abundance erodes the incentive to adapt, magnifying the damage when drought finally outlasts the reservoir. Consistent with this, global demand has outpaced storage for decades, returns of new dams are diminishing (Li et al. 2023), and most reservoirs now fail to refill (Wang et al. 2024). Whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated (Kellner 2021). What the empirical literature still lacks is a counterfactual that separates a reservoir effect from the siting decision without leaning on a modeled unregulated baseline. Paired-gauge comparisons, which read a dam's effect against an unregulated reference reach, have been applied at national scale to flow-regime alteration (Chalise, Sankarasubramanian, and Ruhi 2021, 2023), but not to the transmission of multi-year meteorological drought conditional on the deficit actually experienced, the quantity that determines whether reservoirs buffer the droughts that matter.

We address this with a within-basin falsification test. If a dam buffers drought, streamflow transmission from meteorological to hydrological drought should flatten only below the dam, where regulation operates. If it flattens equally at the unregulated upstream reaches of the same basins, the attenuation is siting, not regulation. This upstream/downstream placebo is the paper's central identification strategy. We complement it with a matched cross-basin comparison that conditions each dammed basin against hydroclimatically comparable undammed basins, and we evaluate both against the drought each basin actually experienced rather than against calendar time, breaking the collinearity between the megadrought onset and the treatment clock.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years shading into structural aridification with a documented anthropogenic component (R. D. Garreaud et al. 2017; R. Garreaud et al. 2025; Zambrano et al. 2025; Boisier et al. 2018; Vicente-Serrano et al. 2019). The 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail, so the policy reflex has been supply-side: new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value export crops (Barría et al. 2021; Reade Malagueño and D'Odorico 2024; Rivera et al. 2016). In the snow-fed basins of central Chile demand already outruns supply in roughly three years of every ten (Webb et al. 2021). Chile thus exhibits every ingredient of the reservoir effect: extreme persistent forcing, a supply-side response, and demand expansion under weak adaptive feedback, making it a most-likely case for detecting a reservoir effect if one operates. At the same time, its short, steep Andean catchments, with streamflow gauges above and below major dams, provide the physical setup the upstream placebo requires, a configuration present across the world's regulated mountain rivers from the Sierra Nevada to the European Alps to the African Rift.

We apply this design across the streamflow, cropland, and water-rights pathways, with an evapotranspiration pathway that we attempt but exclude from the evidence because the available product is biased toward zero over irrigated land (Results). Throughout, outcomes are evaluated at annual to multi-year accumulations, the timescale of structural drought, not of seasonal operation. We find that across the streamflow pathway, over the first decade of the megadrought for which streamflow records exist, the apparent buffering is a siting confound, that the upstream placebo identifies it, and that the operational effect of reservoirs on drought brackets zero once siting is matched out. The upstream placebo is a portable method: wherever rivers have gauges above and below dams, the same falsification test can separate siting from effect.

3 Materials and Methods

3.1 Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyperarid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds, with reservoir presence treated as a basin-level intervention. Because most treated reservoirs (18 of 24) were commissioned before the storage record begins, treatment is effectively time-invariant and a staggered-adoption event study is infeasible; identification rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting* (Di Baldassarre et al. 2018).

3.2 Spatial units and treatment assignment

The analysis grain is the DGA sub-watershed (467 units), chosen over the coarser watershed grain (101 units) because reservoirs sit in Chile's largest main-stem basins and the finer grain supplies more controls inside the treated size range. A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs, giving 24 treated units; after entropy balancing, three high-Andes units fall outside elevation common support, leaving 21 treated basins in all analyses. The reservoirs are predominantly irrigation (20 of 26), so the treated set is dominated by the use type built to buffer agricultural drought, and the streamflow result is unchanged restricted to irrigation-only basins (Supplementary Table S11). An untreated sub-watershed within a reservoir's watershed is flagged as contaminated and removed (45 units), leaving 398 clean controls; this defuses the most direct downstream-spillover (SUTVA) violation. Residual adjacency and shared-aquifer spillovers are addressed with watershed-level checks: where spatial dependence is present, inference is re-run with watershed-block permutation and clustered intervals (Supplementary Table S9), and no matched treated-control pair shares a watershed (Supplementary Table S15).

3.3 Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (the DGA record runs January 2005 to April 2026; we analyse the complete calendar years 2005–2024) with a per-reservoir capacity column (maximum level in hm^3); storage and capacity share units, so percent-of-capacity (`100 × level / maximum level`) is the cross-reservoir-comparable state variable. Static attributes come from the national dam registry.

Meteorological forcing. The Standardized Precipitation-Evapotranspiration Index (SPEI) at a 12-month accumulation, derived from CHIRPS precipitation and CHIRTS temperature and standardized on the 1991–2020 baseline (Sergio M. Vicente-Serrano, Beguería, and López-Moreno 2010), is the primary exogenous dose; CHIRPS-derived indices are validated for Chilean drought monitoring (Zambrano et al. 2017; Meza 2013; Oertel, Meza, and Gironás 2020). The 12-month accumulation matches the annual, snowmelt-fed hydrological cycle of central Chile and so captures the inter-seasonal carryover that irrigation reservoirs bridge; estimates are insensitive to 6- and 24-month alternatives, and the window deliberately does not resolve short-term (1 to 3 month) buffering, which we flag as a boundary of our conclusions (Wu et al. 2017, 2018). Conditioning on forcing, estimating effects on the deficit-to-impact *slope* rather than on levels, breaks the collinearity between the megadrought onset and the treatment clock.

Hydrological drought. Daily streamflow from the DGA network (2000–2020), gathered by the CR2 (Center for Climate and Resilience Research, https://cr2.cl) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12; Sergio M. Vicente-Serrano et al. (2012); Lema et al. (2025)], the regulated-flow analogue of SPEI-12, used as the outcome in the within-basin placebo; gauges are classified upstream or downstream of each dam from SRTM elevation. Because SSI standardizes each gauge on its own, the transmission slope could divide out dam-induced variance compression, so we re-estimate the buffering models and placebo on log 12-month mean flow, a volumetric semi-elasticity no per-gauge normalization can compress, and test the compression premise itself by the deseasonalized relative flow variability by gauge class (Supplementary Table S19).

Ecological and land-use outcomes. (i) annual cropland area from MapBiomas Chile Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum delineated from the CIREN/ODEPA cadastre. Satellite vegetation and evapotranspiration signals track drought-driven losses of agricultural productivity in Chile closely enough to serve as impact outcomes (Zambrano et al. 2018).

Water rights. The DGA consolidated registry of water-use rights, giving per right its grant date, type, use, allocated flow (harmonized to l/s), and capture point. We assign consumptive rights to sub-watersheds and build a per-basin annual panel of the cumulative allocated-rights stock, our direct demand-side outcome. Because the registry contains severe volume errors, the outlier-robust primary measure is rights *density* (count per 100 km^2); a winsorized volume series gives the same result, insensitive to the winsorization threshold (Supplementary Table S8), and rank-based tests on the raw uncapped volumes are reported so no cap or outlier can drive the conclusion (Supplementary Table S18). Both are reported (Supplementary Fig. S6).

Snow water equivalent. ERA5-Land monthly snow-water-equivalent (SWE, 0.1°, 2000–2026), reduced to per-gauge and per-sub-watershed peak-SWE climatologies plus annual peak-SWE anomalies, to test the snowmelt confound on the placebo estimand itself (Supplementary Table S12): we refit the up- and downstream buffering models with a forcing-by-snowpack adjustment, within the low-snow half of units, across early and late panel halves, and with a snow-year interaction. ERA5-Land underestimates absolute Andean snowpack, but the bias is predominantly multiplicative and land-cover-independent, so standardization removes it; all snow terms therefore enter standardized, and gauge-level sensitivity checks among undammed controls are reported per standard deviation of peak SWE. (This differs from the MOD16 ET bias, which is land-cover-dependent and non-multiplicative and therefore disqualifies the ET outcomes, Supplementary Table S21.)

Groundwater levels. DGA well hydrographs of depth to water (2000–2021; 213 of 216 usable wells fall inside the matched basins; a Hampel filter removes spikes), assigned to sub-watersheds by point-in-polygon. Each well's megadrought (2010–2021) depletion rate is the ordinary least squares slope of depth to water on time; per-basin values aggregate well rates by their outlier-robust median. These bound the groundwater-substitution confound as a matched dammed-versus-control ATT on the depletion rate (Supplementary Table S10).

Matching covariates. Baseline aridity (long-term mean annual precipitation divided by potential evapotranspiration, P/PET, over the 1991–2020 World Meteorological Organization normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per sub-watershed by zonal statistics. Because the 1991–2020 normal overlaps megadrought years, we verify the aridity window is innocuous: unit aridity recomputed on the pre-drought 1991–2009 window is essentially identical to the full-window covariate (Pearson $r = 0.998$, Spearman $\rho = 0.999$), so the matching does not lean on drought-period or reservoir-influenced variation (Supplementary Table S13).

3.4 Matched control construction

We estimate the average treatment effect on the treated (ATT) by *entropy balancing*, exact-matched within Köppen main group and balancing *log(area)* and *mean elevation*. Balancing size and elevation rather than latitude is deliberate: latitude and climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size (ESS) from 9 to 91 while balancing latitude as a by-product. The procedure retains 21 of 24 treated sub-watersheds (three high-Andes units fall outside elevation common support) with essentially perfect balance (standardized mean differences for log(area) and elevation fall from 1.18/0.37 to 0.000, variance ratios 1) against 244 in-window controls, the subset of 398 clean controls that survives the common-support trim shared by every matching estimator. Baseline aridity is a genuine confound (median P/PET 0.26 treated versus 0.80 control) but is *not* a balance target, because hard-balancing it collapses the ESS from 91 to 19; the weighting instead reduces the aridity imbalance to a residual SMD 0.17, which the doubly-robust outcome model absorbs. All 21 treated basins lie inside the control aridity range, so the log-aridity adjustment interpolates within overlap rather than extrapolating across it (Supplementary Fig. S8), and the fully non-parametric weighting-only and overlap-subset fits leave the central nulls unchanged (Supplementary Tables S6, S7). The strict hard-balanced fit (also balancing log-aridity, SMD 0.000, ESS 19) is a primary sensitivity: with no extrapolation across the regime gap, even the *apparent* buffering vanishes, the direct signature of an aridity confound (Supplementary Table S17); at ESS 19 it corroborates the powered primary rather than independently powering the result. The decisive evidence, the within-basin placebo, the within-region transmission comparison, and randomization inference, does not use the outcome-model aridity adjustment at all. Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and nearest-neighbour agree on all 21 treated, while CEM prunes the four most extreme basins, and effects are reported with and without them.

3.5 Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline *doubly-robust (DR)* estimator (ebal weights and an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin *transmission slope*, the regression of an annual outcome (e.g. log evapotranspiration) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

Table 1 carries two cross-sectional matched ATTs, both level-change outcomes rather than transmission slopes: (i) *cropland-area expansion*, the change in MapBiomas cropland area per basin over the window (ha km^{-2} added, as Table 1 states), whose protection against differential megadrought exposure comes from the forcing-interacted difference-in-differences below; and (ii) *water-rights accrual*, described under Water rights above. A third cross-sectional outcome, *reservoir ET buffering*, applies the transmission-slope machinery to log evapotranspiration on SPEI over *irrigated orchard cells* (MOD16 ET on the CIREN/ODEPA cadastre footprint); because MOD16 underestimates ET over irrigated land, biasing that slope gap toward zero, the ET outcomes are reported with caveats in the Supplementary Information (Table S21) and carry no evidential weight.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\mathrm{dammed}_i \times \mathrm{SPEI}_{it}) + \gamma\,\mathrm{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with sub-watershed (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect (MDE). Because the headline results are nulls, we bound the operational effect rather than rely on failure-to-reject. For each null we report a two one-sided test (TOST) declaring equivalence when the 90% confidence interval lies within ±25% of the baseline transmission slope, and the threshold-free MDE (80% power, two-sided $\alpha = 0.05$). Because no hydrological standard defines a "negligible" reservoir effect, we anchor the margin in operational drought classification: severity categories are 0.5 standardized units wide (McKee, Doesken, and Kleist 1993; Sergio M. Vicente-Serrano, Beguería, and López-Moreno 2010), and the largest observed SPEI-12 deficit is −2.0, so a 25% slope change perturbs the propagated index by at most 0.29 SSI units, below one severity category. The ±25% margin is therefore deliberately strict, and the qualitative conclusion, that only large effects are excluded and no outcome is equivalent at ±25%, does not depend on the exact bound. The MDE and equivalence interval are built from the *standard deviation of the permutation-null distribution* rather than the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters. A positive control confirms the null is informative: for each $\beta \in \{0, -0.15, -0.30, -0.45\}$ we set $\mathrm{SSI}^*_{it} = \mathrm{SSI}_{it} + \beta \times \mathrm{treat}_i \times \mathrm{SPEI}^c_{it}$, re-fit the estimator and randomization inference, and show recovery is one-to-one in β with detection required once $|\beta|$ exceeds the MDE (Supplementary Fig. S2 and Table S3); the margin translates to a materially non-trivial water volume (median 58 hm³ per 12-month season, Supplementary Table S16), but the causal weight of the null rests on the within-basin placebo.

Siting-confound decomposition. To locate where any apparent buffering comes from, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects); (4) the design estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region. Rung (4), the "regulation estimate" after the between-region gradient is removed, is reported with its own permutation null (Supplementary Table S26). It is important that these are distinct controls. The entropy balancing exact-matches *basin size and elevation* within Köppen main group; it does not balance the residual aridity gradient (which varies within B and C) nor impose a region-specific dose-response. The apparent buffering can therefore survive rungs (1) to (3), whose weighting leaves the within-region aridity gradient intact, and die only at rung (4), which lets each climate region have its own baseline transmission and so removes the between-region contrast that the matching does not. The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated downstream reaches against physically unregulated upstream reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. The contrast is tested directly: the downstream-minus-upstream difference in the differential transmission slope, $D = \beta_{\mathrm{down}} - \beta_{\mathrm{up}}$, is computed on the two gauge-class panels and compared against its distribution under treatment permutation within the same strata (Supplementary Table S25). Because the same treated basins contribute both slopes via their own upstream and downstream gauges, the permutation preserves the paired within-basin structure; a genuine regulation effect would give $D < 0$. To test that the linear transmission specification does not mask a buffering effect confined to extreme deficits, such as threshold-based release rules would produce, we add two checks (Supplementary Tables S28, S29). First, a quadratic SPEI term interacted with treatment on the full panel, so a dry-tail-only effect would appear as a nonzero treatment-by-SPEI-squared coefficient. Second, the transmission models and the placebo contrast are re-fit restricted to meteorological drought months, SPEI below -1.0 (moderate drought and drier) and, as a lower-power sensitivity, below -1.5, each with unit-level permutation inference. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo's comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed. Because the gradient is real, we additionally fit the placebo with the transmission model adjusted for elevation, adding a SPEI-by-elevation interaction so the differential slope is estimated net of the geomorphic gradient, for the ITT, downstream, and upstream panels and the downstream-minus-upstream contrast, each under the design's permutation inference (Supplementary Table S31). We apply the same test to catchment area, since upstream reaches are expected to have smaller drainage areas, whose greater natural storage and longer routing times could independently flatten transmission: we quantify the up/down unit-area gap and regress the per-gauge control transmission slope on log unit area, with a tercile check for non-monotonicity (Supplementary Table S30). The unit area is the sub-watershed area each gauge sits in, a proxy for the drainage feeding it rather than the true upstream catchment. The linearity of that sensitivity is checked across the ~2000 m snowline with a piecewise fit among controls (no detectable break, interaction $p = 0.25$; Supplementary Table S20), and the snow regime itself is tested directly on the placebo estimand (Supplementary Table S12). To exclude that the pooled 2005–2024 slope averages an early-phase buffering (storage intact) into a late-phase failure (storage depleted), the buffering and placebo models are re-fit separately on the early (2010–2014) and late (2015–2024) megadrought phases, and the early-to-late change is tested as a treat $\times$ SPEI $\times$ phase interaction on the pooled megadrought panel (year fixed effects absorb the phase level; Supplementary Table S24). The per-phase subsample refits and the pooled interaction are different specifications, so their point estimates need not reconcile arithmetically; the pooled interaction is the phase-shift test, and its inference, like all small-cluster inference in the design, is by treatment permutation rather than cluster-robust standard errors (Supplementary Table S27). Gauge class (upstream versus downstream) is assigned from SRTM elevation relative to the reservoir outlet (a gauge below the dam elevation is "downstream"). Because a routed stream network is not available in the study data, we bound the misclassification risk two ways (Supplementary Table S32): against the DGA hydrographic sub-cuenca hierarchy, we flag downstream gauges whose sub-cuenca is named for a different river or an unregulated tributary than the dam and re-estimate the placebo without them; and we re-estimate dropping the downstream gauges within 100 m of the dam elevation, the ones most plausibly on a parallel tributary. The geomorphic relief check and the gauge-elevation sensitivity analyses further bound the risk (Supplementary Tables S20, S22); a residual caveat remains that the classification is not validated against routed network topology. Because the local polygon area does not capture the cumulative drainage integrated by a downstream gauge, we test the cumulative-scale confound directly using each gauge's mean annual discharge (proportional to contributing area): the buffering model adds a SPEI-by-log-discharge interaction, so the differential slope is net of cumulative scale, and we report the control-gauge scale sensitivity and the downstream/upstream discharge ratio among paired basins (Supplementary Table S33). The paired placebo's power is quantified as the downstream-minus-upstream contrast MDE at 80% power, built from the permutation-null SD (Supplementary Table S34).

3.6 Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is *randomization inference*: the treatment label is permuted among units within Köppen × aridity-tercile strata (999 permutations). Because outcomes can be spatially autocorrelated, we assess residual dependence with Moran's I (Supplementary Table S9); where it is present, notably the water-rights outcomes ($I \approx 0.4$ to 0.6, $p = 0.001$), we re-run those outcomes with a spatially-restricted permutation that swaps treatment at the watershed-block level, and this spatially-valid inference governs where the naive and spatial tests disagree. Parallel pre-trends are checked with a dynamic event study (dammed × year, reference 2009), and residual confounding is probed with an aridity placebo relabeling the most-arid control tercile as pseudo-treated.

3.7 Software and reproducibility

All analyses are implemented in R within a `targets` pipeline (matched-set construction, estimators, and figures are individual targets), using `WeightIt`/`cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, `sf`/`terra`/`exactextractr` for the spatial extractions, and `renv` for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand. The complete R session information and package versions are archived alongside the pipeline output and reproducible via `renv::restore()`.

4 Results

4.1 Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, so a raw comparison would confuse the reservoir with its location. We separate them with the design in Figure 1. First, we compare each dammed basin only with undammed basins sharing its climate, size and elevation, keeping 21 of 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and S5), and we condition on the drought each basin experienced, comparing impact per unit of deficit. Second, within each dammed basin we compare regulated reaches below the dam against unregulated reaches above it. A genuine reservoir effect must appear in these comparisons; in none does it.

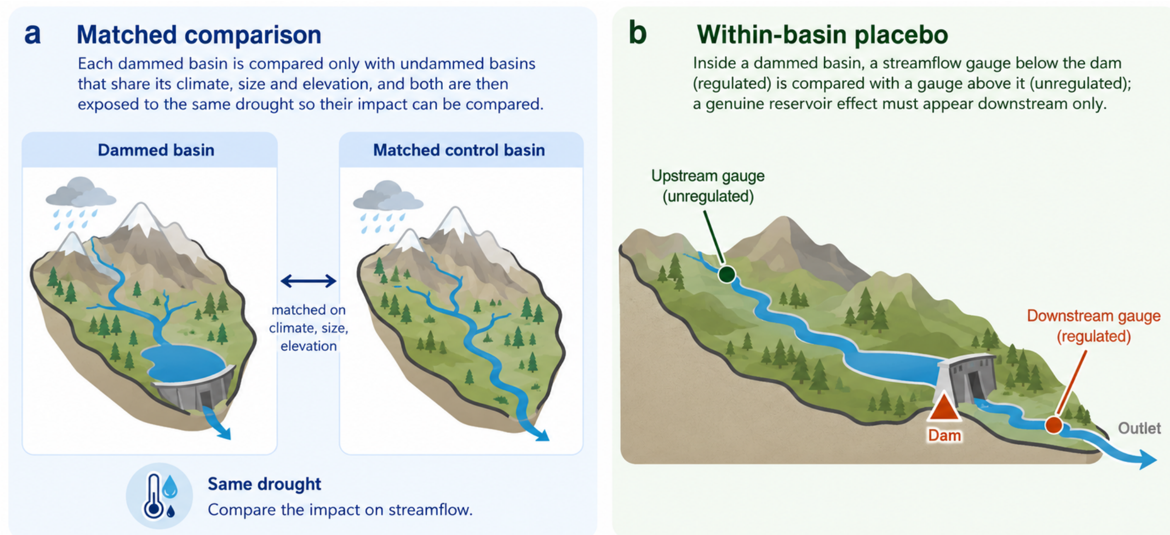

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

4.2 The apparent buffering is the location, not the dam

The most direct test is on streamflow. Meteorological drought (Standardized Precipitation Evapotranspiration Index of 12 months, SPEI-12) propagates into streamflow drought (Standardized Streamflow Index of 12 months, SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope −0.20 upstream versus −0.16 downstream; Figure 2, panel b). A direct within-basin test of this contrast confirms it is null: the downstream-minus-upstream difference in the differential transmission slope is +0.04 with a paired permutation $p = 0.72$, so the regulated reaches show no attenuation beyond the unregulated placebo of the same basins (Supplementary Table S25). The national slope gap of -0.18 accordingly collapses to about -0.05 once dammed and control basins are compared within the same climate region, a residual within its own permutation null ($p = 0.65$; Supplementary Table S26), and in the high-supply south, where buffering should be easiest, the upstream and downstream gaps are identical (-0.036 versus -0.034). The pooled gap does not survive randomization inference, which asks how often reshuffling the dammed label among comparable basins produces a gap this large by chance ($p_{\text{perm}} = 0.39$). Nor is the pooled slope hiding early buffering: at drought onset (2010–2014, storage near-full) the upstream and downstream gaps are identical (-0.43 both). In the late phase the upstream placebo relaxed substantially (to -0.21) while the noisy downstream estimate (standard error 0.36) did not move, but under the design's permutation inference neither phase shift is significant (downstream $p = 0.36$, upstream $p = 0.09$) and the downstream-minus-upstream contrast of the shifts is null ($p = 0.80$; Supplementary Table S27). The cluster-robust signal of differential weakening does not survive the valid small-cluster inference, and the temporal pattern is not distinguishable from the shared forcing regime rather than reservoir depletion. Nor does the linear specification mask a dry-tail effect. A quadratic SPEI term interacted with treatment is permutation-null ($p = 0.65$), so a nonlinear buffering signal concentrated in extreme deficits, as threshold-based release rules would produce, is not detected (Supplementary Table S28). Restricting the transmission models to meteorological drought months (SPEI < -1.0) leaves every buffering estimate and the placebo contrast null (downstream -0.20, $p = 0.71$; upstream placebo $+0.05$, $p = 0.94$; contrast $p = 0.49$), and at the more extreme threshold (SPEI < -1.5) the sample collapses to about 120 unit-months and the estimates become uninformatively noisy (Supplementary Table S29). The apparent buffering is therefore the between-region aridity gradient, the signature of a siting confound, not regulation; a decomposition confirms the signal dies only when each climate region has its own baseline transmission (Supplementary Fig. S3 and Table S4). Hard-balancing aridity gives the same verdict without parametric adjustment: with log-aridity fully balanced (effective control sample 19), the apparent buffering vanishes (intent-to-treat -0.02, upstream -0.05, permutation $p > 0.85$; Supplementary Table S17), exactly what a siting confound predicts. This corroborates but does not by itself carry the result: at an effective control sample of 19, the hard-balanced fit detects only very large effects (an 80%-power MDE near 77% of baseline, versus 46 to 59% in the primary), so its contribution is agreement with the powered design rather than independent power, exactly as its methods note states. The main matched analysis, which retains an aridity residual (standardized mean difference 0.17), is not left confounded: the doubly-robust outcome model absorbs that residual, and the decisive within-basin placebo and the within-region comparison need no aridity balance at all. The parametric adjustment it replaces interpolates within support, every dammed basin inside the control aridity range (Supplementary Fig. S8). Nor is the null a standardization artifact: SSI is normalized per gauge, which would divide out dam-induced variance compression, but on log flow the gaps stay null (downstream $+0.04$, upstream $+0.10$, permutation $p \geq 0.42$) and downstream relative variability is uncompressed ($p = 0.43$; Supplementary Table S19). A direct geomorphic metric agrees: local relief, separating bedrock from alluvial reaches, does not predict control transmission, leaves the placebo unchanged, and shows no buffering among alluvial units, the subsample with the fewest alluvial units being illustrative rather than evidentiary (3 treated upstream and 7 downstream; Supplementary Table S22). Relief adjustment moves the downstream estimate from −0.16 to −0.20, toward the upstream placebo value rather than away from it, which is consistent with the siting interpretation (Supplementary Table S22). Catchment size cannot manufacture the up/down attenuation either: upstream gauges sit in units of essentially identical area to downstream gauges (mean 3,156 versus 3,071 km^2), and among control gauges the transmission slope does not vary significantly with log area (−0.07 per e-fold, $p = 0.06$, $n = 84$), with a non-monotonic tercile pattern (0.87, 0.89, 0.73 from small to large; Supplementary Table S30), so even a weak area sensitivity cannot drive a contrast between equally sized upstream and downstream catchments. Gauge misclassification cannot manufacture the contrast either. Downstream gauges are assigned from SRTM elevation relative to the dam, so a gauge on an unregulated parallel tributary would artificially narrow the down/up contrast. Against the DGA hydrographic hierarchy, 67 of 70 downstream gauges lie in a sub-cuenca named for the dam's own river; three sit on a tributary or different river (two on the Duqueco below the Biobío dams, one on the Itata below the Laja). Excluding those three leaves the contrast null ($D = +0.04$, $p = 0.69$), and excluding the sixteen gauges within 100 m of the dam likewise leaves it null ($D = +0.12$, $p = 0.47$), moving the downstream estimate toward zero rather than toward buffering (Supplementary Table S32).

Cumulative catchment scale cannot manufacture the contrast either. The local polygon area does not capture the drainage integrated by a downstream gauge, but the gauge's own mean discharge, proportional to cumulative contributing area, does, and larger scale genuinely flattens transmission among control gauges (slope +0.08 per log discharge, $p < 0.001$, $n = 84$; Supplementary Table S33). Adding a SPEI-by-log-discharge interaction so the differential slope is net of cumulative scale leaves every effect and the contrast null (intent-to-treat −0.09, $p = 0.50$; downstream −0.10, $p = 0.50$; upstream placebo −0.07, $p = 0.68$; contrast $D = -0.03$, $p = 0.74$; Supplementary Table S33). The scale gap is small in most paired basins, the downstream/upstream discharge ratio below 2 in 10 of 15 (median 1.1), because the downstream gauges sit just below the dams before large tributaries enter. The paired placebo's power is likewise disclosed: with 15 basins it detects at 80% power only a downstream-minus-upstream gap above 0.28 SSI units, about 52% of baseline, so moderate downstream-specific buffering is not excluded (Supplementary Table S34); the placebo's value is discriminating a regulation signature from a siting one, which the elevation-, scale-, and misclassification-robust estimates confirm, not ruling out every buffering magnitude. Upstream gauges sit higher, and the elevation gradient is real: among control gauges, the transmission slope flattens with elevation (−0.21 per km, $p < 0.001$, $n = 84$), and the 0.54 km mean up/down elevation gap would predict a steeper (more negative) upstream differential slope of about 0.11 units. The raw up/down similarity is therefore not decisive on its own, so we test the placebo directly with the transmission model adjusted for elevation: adding a SPEI-by-elevation interaction (so the differential slope is estimated net of the geomorphic gradient) leaves every treatment effect null and the up/down contrast unchanged (intent-to-treat −0.20, $p = 0.30$; downstream −0.20, $p = 0.33$; upstream placebo −0.21, $p = 0.39$; contrast $D = +0.02$, $p = 0.89$; Supplementary Table S31). The elevation adjustment does not reveal a downstream-specific buffering hidden behind the gradient, despite the elevation term itself being strongly significant. The gradient also cannot manufacture a downstream effect in the raw comparison: the 0.54 km gap times the control sensitivity already over-explains the observed -0.04 difference, leaving a residual downstream-specific buffering of at most about 0.07 (13% of the 0.53 downstream baseline), well below the 46 to 59% detectable. A piecewise fit around the Andean snowline (~2000 m) finds no break in this elevation sensitivity ($p = 0.25$; Supplementary Table S20). The placebo's identification therefore does not hinge on an untested elevation assumption: the confound is quantified, adjusted for directly, and it cannot produce the downstream-specific buffering a genuine reservoir effect requires (Supplementary Table S2).

4.3 The powered nulls exclude operational buffering; demand pathways are bounded

The same answer holds across the remaining pathways that carry evidential weight (Figure 3, Table 1). Each effect is the differential response of dammed and matched-control basins to the same deficit, each estimate scaled by its own standard error, so a value inside ± 2 is indistinguishable from no effect. Every estimate that carries evidential weight brackets zero: cross-sectional cropland-area expansion (-0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and its forcing-conditioned slope gap ($p_{\mathrm{perm}} = 0.91$). The evapotranspiration pathway carries no evidential weight and is excluded rather than counted as a null: whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the post-hoc orchard stratum relies on MODIS (Moderate Resolution Imaging Spectroradiometer) MOD16 product, whose documented bias toward zero over irrigated land, a product property identified independently of our results, disqualifies its null (Supplementary Table S21, Fig. S7).

The null is informative but its power is concentrated in the streamflow pathway. With about 21 dammed clusters the design cannot claim exact equivalence to zero but rules out operational buffering above 46 to 59% of baseline, and a positive control confirms it: injected buffering is recovered one-to-one and detected beyond that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). This is a wide null: moderate buffering (e.g., a 30 to 40% reduction in transmission) is fully compatible with the data; only very large effects are excluded, so the finding is no *large* operational buffering, not none. The powered streamflow null covers the record that ends in March 2020, so it spans the first decade of the megadrought but not the 2019–2022 peak deficits; the storage and cropland pathways, which run to 2024, are the only multi-year-drought evidence for the late phase. Of the remaining pathways, only the streamflow null excludes material effects at operationally relevant magnitudes. The cropland and evapotranspiration designs are far weaker, with minimum detectable effects far above baseline (7,330% and 237% of baseline, respectively; Supplementary Table S2); their nulls corroborate rather than independently establish the absence of an effect. The water-rights count is comfortably null under all inference frameworks ($p_{\mathrm{perm}} \geq 0.42$), while the winsorized volume is a more nuanced result discussed below.

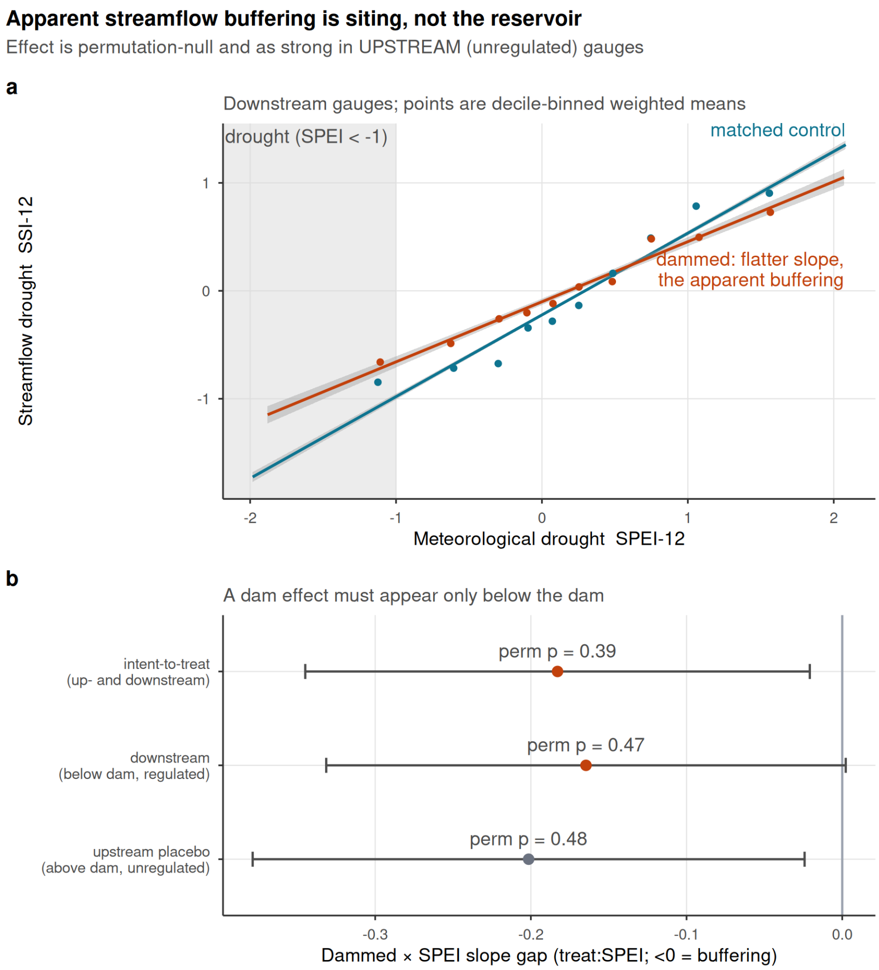

Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD row and water-rights accrual are judged by randomization-inference p-values (p), the design's primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21, Fig. S7): whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the orchard rows rely on MOD16, whose documented underestimation over irrigated land biases them toward zero.

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km^2 added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null

4.4 Reservoirs mark prior vulnerability; megadrought divergence is bounded, not absent

Dammed basins carry the vulnerability reservoirs are blamed for concentrating: about 4.5 times more cropland than matched controls (10.8% versus 2.4% of basin area; Figure 4, panel a). This gap is a *siting* signature, stable across the 2000–2024 record. A dynamic event study finds the differential trend flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) with no post-2010 divergence detected ($p = 0.81$; Figure 4, panel b). Both are bounded nulls, not confirmations of parallelism: the pre-trend test detects only divergence accumulating to about half the control cropland level by 2009 (the observed trend is a seventh of that), and the permutation envelope widens after 2015, so the late panel cannot rule out substantial divergence (Supplementary Table S14). The null is metric-invariant: on log cropland share the slope gap is likewise permutation-null ($p = 0.37$; Supplementary Table S20). Reservoirs mark where cropland already was. The absence of detected divergence is a bounded null, not a confirmed absence: the design cannot rule out late-panel divergence comparable to the control cropland level, so within these bounds the record is consistent with no megadrought-era expansion, but substantial divergence in the later years remains possible.

The same holds for the most direct demand measure. In the Chilean Water Directorate (DGA) registry of consumptive water rights, dammed basins appear to accrue rights faster than controls, on both the outlier-robust count (+20 per 100 km²; Figure 3, Table 1) and the winsorized volume (+2.5 l/s per km²; Supplementary Fig. S6), each with an analytic interval excluding zero. This is the naive-analysis trap the design catches: dammed basins are more arid, arid basins carry denser water-rights activity, and the imbalance drives the gap. The count does not survive randomization inference ($p_{\text{perm}} = 0.52$; Supplementary Fig. S6) and remains null under a fully non-parametric match on the aridity-overlap subset ($p_{\text{perm}} = 0.42$; Supplementary Table

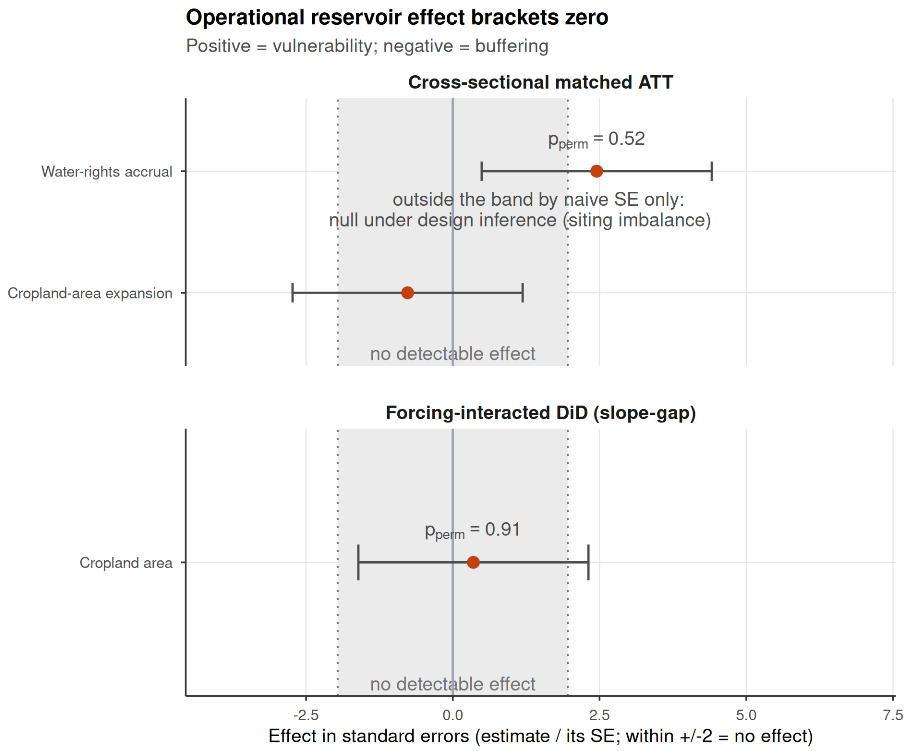

Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span +/-1.96 SE (the 95% reference interval) and dotted lines mark +/-1.96; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. Rows judged by randomization inference are annotated with their p (the design's valid inference at ~21 clusters, since the +/-1.96 SE guides over-reject there). Under that valid inference every estimate brackets zero. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21) as product-limited and confounded. Water-rights accrual sits outside +/-1.96 by its naive standard error yet is null by the design (p_perm = 0.52; the apparent induced demand lies inside the permutation null expected from the dammed basins' unbalanced aridity, Supplementary Fig. S6).

S7). The winsorized volume is more delicate: stable across thresholds and physically validated (Supplementary Table S8). On the full matched sample the volume null is insensitive to the winsorization threshold (permutation p from 0.39 to 0.21 across the 95th to 99.9th percentiles), while on the aridity-overlap subset, a smaller sample of common-support units, the unit-level permutation p falls to 0.005; the residuals are spatially autocorrelated (Supplementary Table S9) and under the governing cuenca-block permutation it is not significant ($p_{\text{perm}} = 0.077$, cuenca-clustered interval $[-0.3, 4.1]$). Ranked raw, uncapped volumes give the same verdict: the design rank contrast is null under cuenca-block permutation ($p = 0.15$; Supplementary Table S18). The water-rights evidence is therefore not uniformly null: the count is cleanly so under all frameworks, the ranked raw volumes are null under spatial inference, and the winsorized volume sits closer to the boundary, with a unit-level signal ($p = 0.005$) that the spatial structure partially explains but does not fully erase (cuenca-block $p = 0.077$). The gap between naive and spatial inference flags the spatial concentration of large consumptive rights in dammed cuencas, a pattern the design can bound but whose attribution to siting versus reservoir-induced demand it cannot fully resolve at this spatial grain. No induced demand is robustly attributable to the reservoir, as measured by the accrual of new state-granted rights, once siting and spatial structure are accounted for; but the volume signal warrants further investigation with parcel-level consumption data. The measure covers grants, not the private market reallocation of existing rights, which in Chile's administratively closed basins is the channel through which rights actually change hands and which this registry does not record (Discussion), so no conclusion is drawn about market-driven concentration.

4.5 Storage falls; the carryover buffer shrinks

If the streamflow pathway excludes material operational buffering and the demand-side pathways are consistent with no amplification at their respective detection limits, what matters is how much water there is to store (Figure 5). Over 2005–2024 the storage band drifts downward: peak storage falls by 1.3 percentage points of capacity per year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1}, $p < 10^{-4}$), while the seasonal amplitude shows no detectable trend (-0.06 pp yr^{-1}, $p = 0.91$; Supplementary Table S5). The unchanged amplitude describes the within-year refill cycle, which is not the buffering mechanism that matters here. For drought on the multi-year timescale this paper targets, the buffering mechanism is carryover volume, the water held across years, and the storage level decline directly shrinks it: peak storage fell from 87% of capacity before 2010 to 67% after, and the trough from 40% to 25% (Supplementary Table S5), so the pool available to buffer a second and third dry year is materially smaller. This is a real reduction in multi-year buffering capacity. Whether the decline is supply-driven or management-driven cannot be attributed causally: the trend is descriptive, has no control group, and is confounded with potential operational or management changes, so it supports no causal attribution. It is, however, directionally consistent with a supply-side reading, in which the fall tracks the decline in unregulated inflow rather than a failure of the reservoirs to refill to their seasonal pattern, which the unchanged amplitude indicates they still do when water arrives. The direction is consistent with the supply reduction measured independently in the

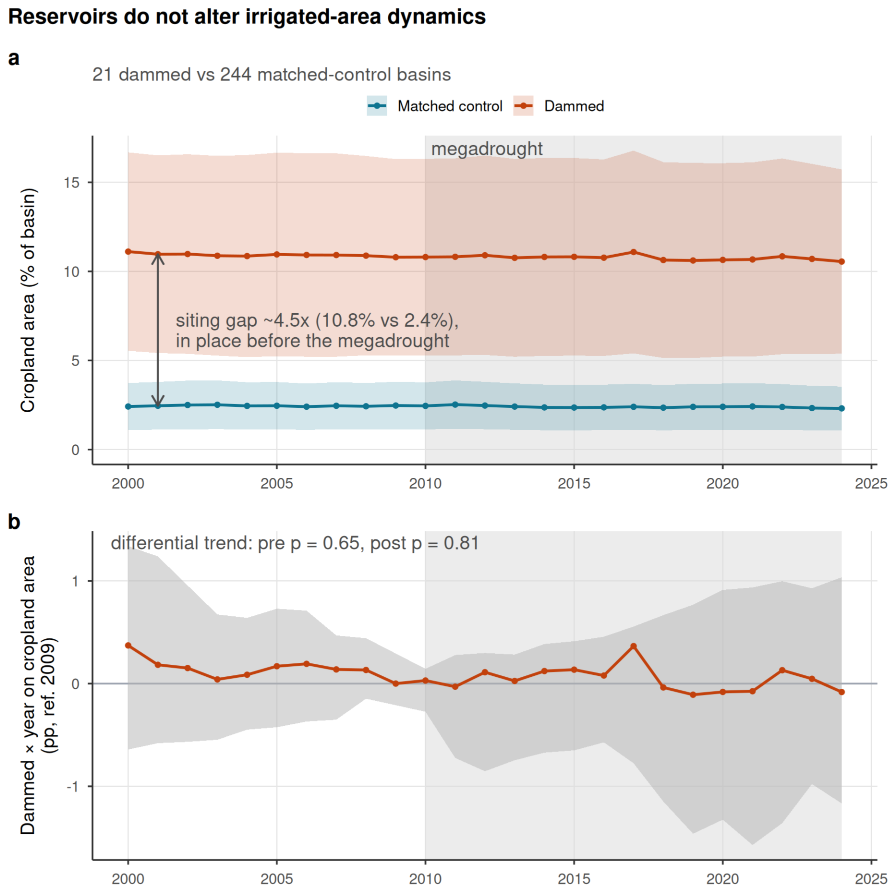

Figure 4: Observed cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean +/- 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold ~4.5x more cropland (10.8% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009): the grey band is the 95% randomization-inference null envelope (2.5th to 97.5th percentile of the dammed x year coefficients under treatment permutation within Köppen x aridity strata); coefficients inside the band are indistinguishable from no divergence, and cluster-robust intervals are omitted as anti-conservative at ~21 treated clusters. The differential pre-trend is flat (pre-2010 trend-slope cluster-robust $p = 0.65$, permutation $p = 0.72$) and no post-2010 divergence is detected (post $p = 0.81$); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~21 treated clusters. The null envelope widens after 2015, reflecting declining power in the later panel, so the flat post-2015 trajectory is a bounded null: divergence smaller than the envelope cannot be ruled out. The 2010-2024 megadrought is shaded in both panels.

matched controls, though the magnitudes differ: unregulated control streamflow fell about 2.9% per year over 2005–2020 (dammed 3.4%), a flux decline larger than the 1.3 pp yr^{-1} fall in the storage stock, as expected since storage is a buffer that partially absorbs inflow shocks, and the runoff deficit documented for the megadrought (R. D. Garreaud et al. 2017; Alvarez-Garreton et al. 2021) is an order of magnitude larger in percentage terms. Storage is a mass balance, so the band alone cannot separate falling inflows from sustained releases; the supply reading rests on that independent streamflow decline. Our direct demand outcomes point the same way: neither cropland area nor water-rights allocation shows expansion (Figure 4, Table 1; Supplementary Fig. S6) (Li et al. 2023; Wang et al. 2024).

5 Discussion

Across a matched national design, multiple estimators, and a within-basin falsification test, the apparent drought buffering attributed to reservoirs is a measurable siting confound. The upstream placebo is the decisive evidence: a regulation effect must appear only below the dam; it does not, and what remains is the aridity gradient. Dammed basins hold about 4.5 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no megadrought divergence detected within the design's power (a bounded null, not a confirmed absence). One reading fits: reservoirs mark where water-dependent vulnerability already concentrated rather than make it. This is a finding, not a failure of power: the apparent effect has an identified cause and the positive control shows the design detects buffering when it is present.

The corollary is transferable: dammed-versus-undammed or before-after contrasts inherit the full siting confound (López-Moreno et al. 2009; Wu et al. 2017; Schilstra et al. 2024; Lee, Kang, and Kim 2026). The upstream placebo ports to regulated rivers with gauges above and below a dam, most directly to short, steep catchments like Chile's; in large low-gradient basins, floodplain storage and transmission losses would blur the contrast; porting it would need routing-corrected lags or naturalized-flow counterfactuals.

If the streamflow pathway excludes material operational buffering at multi-year scale and the demand-side outcomes are consistent with no amplification at their respective detection limits, what matters is the water available to store. The storage band steps down at the megadrought onset, peak and trough together, which directly shrinks the carryover volume held across years, the buffering mechanism for multi-year drought: the pool available to carry a second and third dry year is materially smaller by 2024 than it was in 2010. This is a real reduction in multi-year buffering capacity. Whether it is supply-driven or management-driven cannot be attributed causally, since the trend is descriptive and confounded with operational change; it is directionally consistent with a supply-side reading. The seasonal amplitude is unchanged within a wide interval, indicating that the reservoirs still refill to their seasonal pattern when water arrives, consistent with the carryover loss tracking falling inflow rather than a failure of the refill cycle. The powered result is the level decline, the expected mass-balance

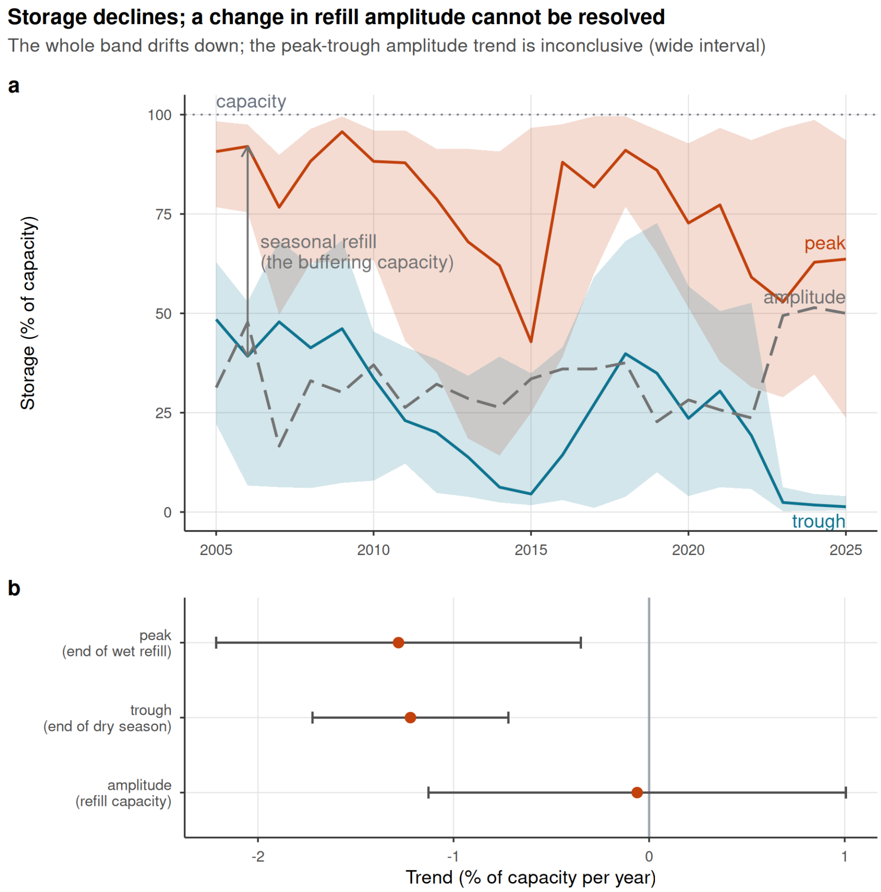

Figure 5: Reservoir storage declines; the carryover buffer shrinks with supply. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant). The level decline shrinks the carryover water held across years, the multi-year buffering mechanism; the unchanged amplitude shows the within-year refill cycle is intact, so the carryover loss tracks falling inflow, a supply-side decline, rather than a failure to refill seasonally. Descriptive: pooled within-reservoir trend, no control group.

outcome of persistent inflow deficits; descriptive and confounded with management change, it supports no causal attribution but is directionally consistent with the independent decline in unregulated control streamflow, which fell faster (2.9% per year) than the storage stock (1.3 percentage points of capacity per year), as a buffer should partially absorb an inflow shock. Read against Chile's structural aridification (Zambrano et al. 2025; R. D. Garreaud et al. 2017; Alvarez-Garreton et al. 2021), accelerating Andean warming and snow loss now cutting into the snow-fed streamflow these basins depend on (Kim, Rajaram, and Lakshmi 2025), and the global pattern of reservoirs failing to refill (Wang et al. 2024; Li et al. 2023), the binding constraint is shifting toward supply.

These findings give the contested socio-hydrological *reservoir effect* and *supply-demand cycle* (Di Baldassarre et al. 2018; Kellner 2021) a credibly identified test, and find the protective function over-attributed to storage. Because treatment is time-invariant, we test the spatial imprint of these feedbacks, not the longitudinal loop. The 4.5-fold cropland gap could be the equilibrated legacy of demand induced at construction, so our null covers the megadrought era; historical induced demand is testable only with staggered commissioning. And the record opens in the scarcity phase, after any expansion under perceived abundance, so it bounds drought-era expansion, not the cycle as a whole. Within scope the nulls refine the theory: the cycle leaves no operational imprint during drought, so its mechanisms likely act through siting and construction rather than operation, a distinction current models omit. How far the demand-side null travels depends on governance: the siting lesson holds wherever dams are placed non-randomly; prior appropriation, which forfeits unused rights, may amplify induced demand while adaptive allocation curtails it. Whether the null is general or the mark of a fully allocated market we cannot adjudicate: rights kept accruing in dammed basins, but systems with water still unallocated could express an induced demand that ours cannot. The upstream placebo itself ports more broadly. Short, steep catchments with gauges above and below dams are common across the world's regulated mountain rivers: the California Coast Ranges and Sierra Nevada, the Spanish Sierra Nevada and Pyrenees, the Italian Apennines and Alps, the African Rift, and the Andean cordillera beyond Chile. In large low-gradient basins, the Murray-Darling, the Colorado below Lake Mead, or the Indus plains, floodplain storage and transmission losses would blur the above/below contrast and routing-corrected lags or naturalized-flow counterfactuals would be needed. The placebo method is therefore not universal, but its scope conditions are explicit and its required data, paired gauges above and below dams, exist across much of the world's irrigated mountain-front hydrology.

Chile's 1981 Water Code makes water-use rights perpetual private property, so the reflexive response has been supply-side, new dams and inter-basin transfers (Barría et al. 2021; R. Garreaud et al. 2025); the 2022 reform moves toward the adaptive allocation our results argue for (Macpherson et al. 2023) but postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering dams provide is a property of where they sit, not of what they do (Ho et al. 2017). Adaptation must shift toward demand management, reallocation and managed aquifer recharge (Grafton et al. 2018; Scanlon et al. 2023; Rivera-Vidal et al. 2025).

Several limitations bound these conclusions, each quantified in the Supplementary Information. The result is a bounded null: the design excludes streamflow buffering above 46 to 59% of baseline, demand-side outcomes are far less powered and the pre-trend test power-limited (Supplementary Tables S2, S14), so causal weight rests on the placebo, demand nulls corroborating; all survive the hard-balanced match (Supplementary Table S17). The powered streamflow null also ends in March 2020, before the most acute 2019–2022 deficits, so the late-phase storage and cropland pathways carry that part of the record. The early-to-late phase analysis is likewise null under the design's permutation inference, with neither the downstream nor the upstream phase shift significant and their contrast far from it (Supplementary Table S27), so the flat late-phase downstream estimate is a noisy (standard error 0.36) point within its null rather than evidence of regulation; cluster-robust intervals, which the design's own rule says over-reject at about 21 clusters, overstate the phase signal. The registry measures paper rights, not wet water: Chilean basins are widely over-allocated and granted rights can sit unused or hoarded (Barría et al. 2021; Reade Malagueño and D'Odorico 2024), so under drought, when physical availability binds, the accrual we test is an entitlement response that non-use could dampen; cropland is the physical check. The registry also records original grant dates, not market transfers: many basins have been administratively closed to new allocations for decades, so the flat accrual reflects the absence of new state grants rather than the reallocation of existing rights, which in Chile's private water market proceeds through transactions registered in local real-estate offices, not the DGA grant registry (Reade Malagueño and D'Odorico 2024). Market-driven concentration of rights below reservoirs, if present, is therefore invisible to the accrual measure; cropland trajectories are the physical check that argues against such concentration at scale, but a transaction-level test would be needed to close this pathway. Both metrics also miss intensification within an unchanged footprint; central Chile's documented shift toward thirsty export fruit (Reade Malagueño and D'Odorico 2024) is exactly such a transition, invisible to both; a consumption-grade test awaits thermal ET or crop-transition maps. Groundwater substitution is bounded, dammed and control water tables falling at indistinguishable rates across 213 DGA wells despite severe regional drawdown (Taucare et al. 2024); monitored wells need not sample the orchard footprints where pumping concentrates, so unmonitored drawdown could sustain a historically reservoir-induced footprint (Supplementary Table S10). In that reading the reservoir effect is not absent but displaced to the aquifer, a construction-era legacy persisting on a pool the design cannot see; the well bound and flat cropland trajectories argue against substitution at scale but do not close the pathway. The snowmelt confound, acute in these snow-fed Andean basins (Ayala et al. 2025; Götte and Brunner 2024), is tested on the differential estimand and excluded across four specifications (Supplementary Table S12). The geomorphic check leaves the placebo unchanged, though 2.5 km relief cannot resolve aquifer storage itself, a residual caveat (Supplementary Table S22). The placebo also spans land-cover and abstraction contrasts, downstream valleys more cultivated, upstream reaches carrying unrecorded diversions; gauge fixed effects absorb these level differences, and drought-intensified upstream abstraction would steepen upstream transmission, working against the observed upstream attenuation, not producing it. Administrative closure cannot force the demand null: treated basins entered the drought with higher allocated stock yet accrued rights 2.7 times faster, and parallel control cropland contradicts collapse prevention (Supplementary Table

S15). No matched pair shares a watershed; outbound transfers would shrink the gap and imports would mimic downstream-specific buffering, which is absent (Supplementary Table S15); irrigation-only restriction leaves the result unchanged (Supplementary Table S11). MOD16's bias toward the null excludes the ET pathway from evidence (Supplementary Table S21); the demand conclusion rests on cropland and the registry, with thermal ET as future work. Finally, the 12-month accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs also provide, which we neither test nor dispute. The fleet is partly seasonal by design, the median treated reservoir storing 0.3 years of downstream flow, yet seven of 17 hold at least half a year, four exceed a year, and none empties seasonally (Supplementary Table S23): carryover was physically available, so its absence in outcomes is not a design tautology. Two extensions can test what our window cannot: a staggered event study of cropland around historical commissioning would probe construction-phase induced demand; a rule-curve quasi-experiment on release records, once public, would isolate operation.

6 Conclusions

At the multi-year timescale of structural aridification, then, the design detects no operational buffering above 46 to 59% of baseline and no megadrought-era demand expansion beyond the siting legacy: the apparent buffering is a statistical shadow of where dams sit, not of what they do, while the water available to store declines. The nulls are bounds, not zeros; effects inside them can be volumetrically material (Supplementary Table S16). But under structural aridification with no operational buffering within the design's bounds at the multi-year scale, building more storage is a supply-blind response to a supply problem. Chile continues to plan new reservoir construction to expand storage, while its agricultural water demand continues to grow under perpetual private water rights that the 2022 Water Code reform only begins to address (Macpherson et al. 2023). The alternative is not theoretical: demand reallocation through active water markets, irrigation efficiency, managed aquifer recharge, and crop transitions toward lower water footprints are available now and do not depend on rainfall (Grafton et al. 2018; Scanlon et al. 2023; Rivera-Vidal et al. 2025). Resilience to a falling supply ceiling will come from confronting demand and allocation, not from more impoundments.

7 Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.21052149). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomas Chile (land cover), the CR2 network (streamflow), ERA5-Land (snow water equivalent), SRTM (elevation), and the DGA reservoir, watershed, water-rights, and well-hydrograph registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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